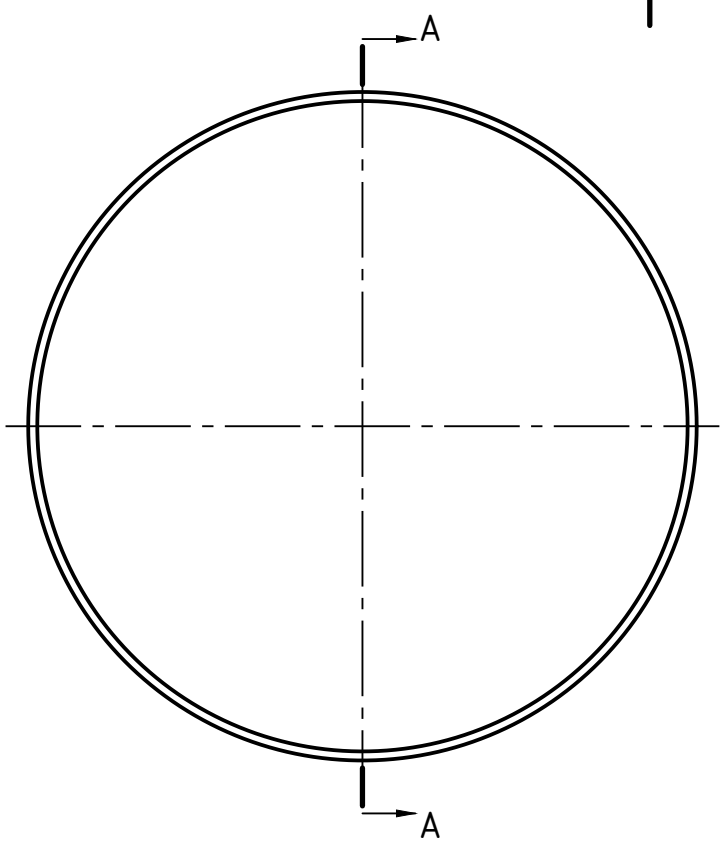
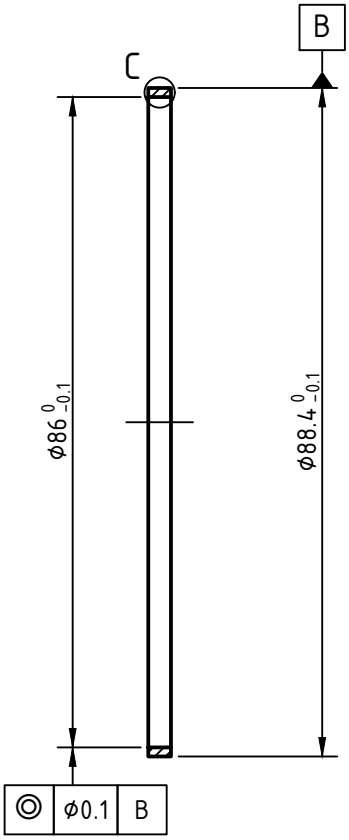
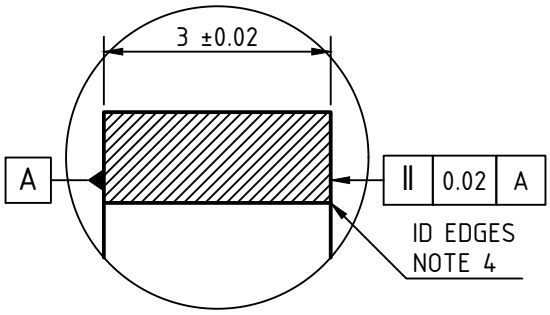




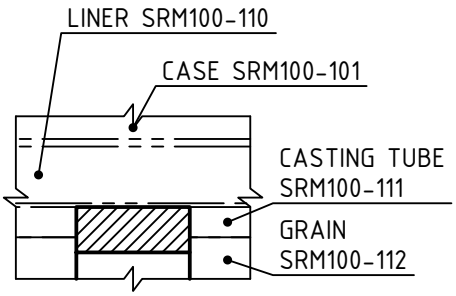
SECTION A-A

NOTES

1. Material (certified): linen phenolic, NEMA LI 1 grade LE (IEC 60893-3-4 PF CC 203 or equal).
2. Stock: sheet 4 (5/32 in), surface-ground equally on both faces to 3 ±0.02 before profiling. Machine dry, no coolant.
3. Cut to grain_spacer.dxf (1:1, mm, layer CUT) by CNC router (carbide compression cutter) or turn a clamped stack on a mandrel; no laser, water jet or punching. The DXF sizes are upper limits: aim ϕ88.35 / ϕ85.95.
4. Edges: ISO 13715 -0.2, but both ID edges burr-free, -0.05 max: the ID band bears on the grain face.
5. Fit: OD in liner SRM100-110 bore ϕ88.6 +0.1/0, diametral clearance 0.20 … 0.40, whatever the case bore (ϕ92.000 … 92.10). The ID covers 0.40 … 0.45 of each grain face (ϕ86.8, radial, concentric).
6. Inspect 100 %: t at 4 points (micrometer), mean on the bag label; OD and ID at 2 × 90°; parallelism on a surface plate, face A down.
7. Replace after every firing. Bag label SRM100-109, batch, mean t; store sealed, with desiccant.
8. USED ON: SRM100-000 motor, item 11 (5 off); not fitted in SRM100-001.



C (10:1)



INSTALLED (5:1) - REFERENCE

Ra 3.2

MATERIAL	MASS	GENERAL TOLERANCES	SURFACE TEXTURE	EDGES
Linen phenolic, NEMA LI 1 LE sheet 4, faces ground to 3	1.3 g	ISO 2768-mK	Ra 3.2	ISO 13715 -0.2, NOTE 4

DOCUMENT TYPE	DOCUMENT STATUS	SCALE	PROJECTION	UNITS
DETAIL DRAWING	NOT CHECKED - FOR REVIEW	1:1		mm

LEGAL OWNER	TITLE, SUPPLEMENTARY TITLE
HELIOS SRM-100 SOLID ROCKET MOTOR	GRAIN SPACER Grain segment spacer - linen phenolic, 5 off per motor

CREATED BY	APPROVED BY	DATE OF ISSUE	IDENTIFICATION NUMBER	REV	SHEET
text-to-cad / Claude	-	2026-09-23	SRM100-109	A	1 / 1